

Q.1(a) $\nabla \cdot \vec{D} = \rho_f$

$\Rightarrow \oint d\vec{s} \cdot \vec{D} = \iiint \rho_f dv$

In the limit $dh \cdot 2\pi dr \ll dA$

$\oint \vec{D} \cdot d\vec{s} = (\vec{D}_2 \cdot \hat{n}_{12} - \vec{D}_1 \cdot \hat{n}_{12}) dA = \rho_f \cdot dA \cdot dh$

$\Rightarrow (\vec{D}_2 - \vec{D}_1) \cdot \hat{n}_{12} = \rho_f \cdot dh = \sigma_s^f$ (by definition)

$\therefore (\vec{D}_2 - \vec{D}_1) \cdot \hat{n}_{12} = \sigma_s^f$

(b) similarly $(\vec{B}_2 - \vec{B}_1) \cdot \hat{n}_{12} = 0$ ($\because$ there is no magnetic charge)

(d) $\nabla \times \vec{H} = \vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t}$

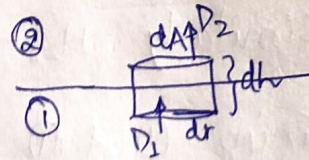
$\Rightarrow \int \nabla \times \vec{H} \cdot d\vec{s} = \int (\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t}) \cdot d\vec{s}$

$\Rightarrow \oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{s} + \frac{\partial}{\partial t} \int \epsilon \vec{E} \cdot d\vec{s}$

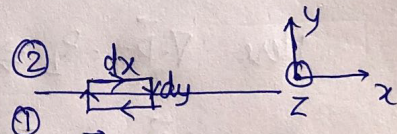
$\Rightarrow (H_{2x} - H_{1x}) dx = -J_z dx dy - \frac{\partial (\epsilon E_z)}{\partial t} dx dy$

($\because \oint \vec{H} \cdot d\vec{l} \approx H_{2x} dx - H_{1x} dx$ (in the limit $dy \ll dx$)

$\Rightarrow H_{2x} - H_{1x} = \underbrace{-J_z dy}_{j_{sz} \text{ (by defn of surface current density)}} - \frac{\partial (\epsilon E_z)}{\partial t} dy$ ($\rightarrow 0$ as $dy \rightarrow 0$ for finite $\vec{E}$ field)



($\because \iiint \rho_f \cdot \vec{D} dv = \oint \vec{D} \cdot d\vec{s}$)



②

$$\therefore H_{2x} - H_{1x} = -\dot{J}_{sz} \quad \text{--- (1)}$$

$$\text{Similarly } H_{2z} - H_{1z} = \dot{J}_{sx} \quad \text{--- (1)}$$

Combining (1) & (1)

$$\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{J}_s$$

$$\text{or } \boxed{\hat{n}_{12} \times (\vec{H}_2 - \vec{H}_1) = \vec{J}_s}$$

(c) similar to part (d), one can show that

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \boxed{\hat{n}_{12} \times (\vec{E}_2 - \vec{E}_1) = 0}$$

Q.2. As used in Q.1, the integral form of Maxwell's equations are

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{f, \text{enclosed}} \quad \oint \vec{E} \cdot d\vec{l} = -\iint j\omega \mu \vec{H} \cdot d\vec{s}$$

$$\oint \vec{B} \cdot d\vec{s} = 0 \quad \oint \vec{H} \cdot d\vec{l} = I_f + \iint j\omega e \vec{E} \cdot d\vec{s}$$

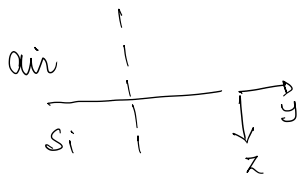
Q.3. If $(\vec{E}(\vec{r}), \vec{H}(\vec{r}))$ satisfy Maxwell's equations in the medium $(\epsilon_1(r), \mu_1(r))$ at frequency ω , then they satisfy Maxwell's equations in medium $(s\epsilon_1(\vec{r}), s\mu_1(\vec{r}))$ at frequency $s\omega$.

$$\begin{aligned}
 \text{Q.5. } \nabla \cdot (\vec{E} \times \vec{H}^*) &= \vec{H}^* \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H}^*) \\
 &= \vec{H}^* \cdot (-j\omega\mu\vec{H}) - \vec{E} \cdot (j\omega\epsilon\vec{E} + \vec{J})^* \\
 &= -j\omega\mu|\vec{H}|^2 + j\omega\epsilon|\vec{E}|^2 - \vec{E} \cdot \vec{J}^*
 \end{aligned}$$

$$\Rightarrow \boxed{\nabla \cdot (\vec{E} \times \vec{H}^*) + j\omega[\mu\vec{H} \cdot \vec{H}^* - \epsilon\vec{E} \cdot \vec{E}^*] + \vec{E} \cdot \vec{J}^* = 0}$$

Q.6. - As we have normal incidence, the results would be identical for TE & TM polarizations.

Let's consider $\hat{x}$ -polarized light



$$Z_a = \sqrt{\frac{\mu_0}{\epsilon_0}}, \quad Z_{si} = \sqrt{\frac{\mu_0}{\epsilon_{si}}} = \sqrt{\frac{\mu_0}{\epsilon_r - j\epsilon_i}}$$

$$r = \frac{Z_{si} - Z_a}{Z_{si} + Z_a}$$

$$t = \frac{2Z_{si}}{Z_{si} + Z_a}$$

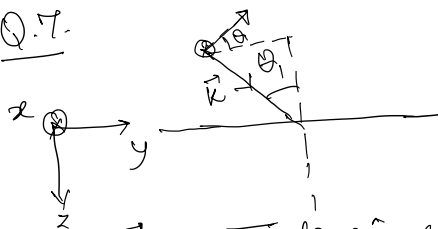
$$K_a = \frac{2\pi}{\lambda_0}, \quad \lambda_0 = 400 \text{ nm}$$

$$K_{si} = \frac{2\pi}{\lambda_0} \sqrt{\frac{\epsilon_r - j\epsilon_i}{\epsilon_0}}$$

$$\vec{E}(\vec{r}) = \hat{x} E_{inc} \times \begin{cases} e^{-jk_a z} + r e^{jk_a z} \\ t e^{-jk_{si} z} \end{cases}$$

$$\vec{H}(\vec{r}) = \hat{y} E_{inc} \times \begin{cases} (e^{-jk_a z} - r e^{jk_a z}) \cdot \frac{1}{Z_a} \\ t e^{-jk_{si} z} \cdot \frac{1}{Z_{si}} \end{cases}$$

Q.7.



$$\vec{k} = \omega \sqrt{\mu_0 \epsilon_1} (\cos \theta_1 \hat{z} + \sin \theta_1 \hat{y})$$

$$\vec{E}_{inc}(\vec{r}) = E_0 (\hat{x} + j(\cos \theta_1 \hat{y} - \sin \theta_1 \hat{z})) \times e^{-j\vec{k} \cdot \vec{r}}$$

Decompose $\vec{E}_{inc}$ into TE & TM polarized components & use superposition principle.